NIOS Class 10 (212) Science & Technology — Guaranteed Sure-Shot Questions & Diagrams

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

NIOS Class 10 (212) Science & Technology — Guaranteed Sure-Shot Questions & Diagrams

> **These 6 question templates appear with near-100% certainty in NIOS Science & Technology papers.**
> Mastering these 6 templates locks in **25 to 30 marks** in Section B! Practice drawing the diagrams and writing the equations with a pen.

Template 1: Human Eye Defects — Myopia vs Hypermetropia (5 Marks — Q43/Q44)

Problem:

> What is Myopia? State its two causes. How can this defect be corrected? Draw ray diagrams showing:
> (i) a myopic eye, and
> (ii) its correction using an appropriate lens.

Complete Model Answer:

1. **Definition**:
2. **Myopia (Short-sightedness)** is a defect of vision in which a person can see nearby objects distinctly, but cannot see distant objects clearly. The far point of a myopic person is nearer than infinity.

3. **Causes**:
4. Excessive curvature of the eye lens (lens becomes too thick / convergent).
5. Elongation of the eyeball.
6. **Nature of Image & Correction**:
7. Light rays coming from distant objects focus **in front of the retina** instead of on the retina.
8. **Correction**: It is corrected by using a **Concave Lens** (diverging lens) of suitable focal length. The concave lens diverges the parallel rays so that they appear to come from the person's far point, allowing the eye lens to focus the image exactly on the retina.
9. **Ray Diagrams (Representation)**:

```
(i) MYOPIC EYE (Image formed in FRONT of retina):
Parallel rays ---------\
                        \--\
----------------------------[ Eye Lens ]-----> (Focuses HERE) ----- [ Retina ]
                        /--/                   (Blurry on Retina)
Parallel rays ---------/

(ii) CORRECTION USING CONCAVE LENS:
Parallel rays -----\
                    [ Concave ] -------\
--------------------[  Lens   ]---------[ Eye Lens ]----------------> [ Retina ]
                    [ (Diverges)]------/                              (Sharp Focus!)
Parallel rays -----/
```

*(Note for Hypermetropia / Long-sightedness: Image forms BEHIND the retina due to eyeball being too short or lens being too thin. It is corrected using a **Convex Lens**).*

Template 2: Ohm's Law Verification & Equivalent Resistance (5 Marks — Q43)

Problem:

> (a) State Ohm's law. Draw a schematic circuit diagram to verify Ohm's law in the laboratory.
> (b) Three resistors of resistances $2\,\Omega$, $3\,\Omega$, and $6\,\Omega$ are connected. How can they be connected to give a total resistance of:
> (i) $4\,\Omega$, and (ii) $1\,\Omega$?

Complete Model Answer:

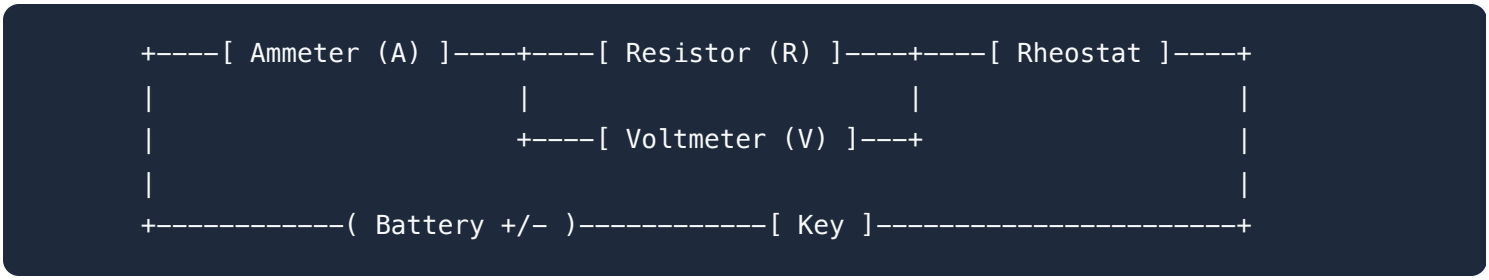
1. **Statement of Ohm's Law**:
2. Ohm's law states that the electric current (I) flowing through a metallic conductor is directly proportional to the potential difference (V) applied across its ends, provided temperature and other physical conditions remain constant:

$$V \propto I \implies V = IR$$

3. *(where R is the electrical resistance of the conductor)*.

4. **Circuit Diagram**:

5. Components connected in series: Battery (cell), Plug key, Ammeter (A), Rheostat (variable resistor).
6. Component connected in parallel: Voltmeter (V) connected across the resistor (R).



1. **Numerical Solution**:
2. **(i) To get $4\,\Omega$**:

- Connect $3\,\Omega$ and $6\,\Omega$ in **parallel**, and then connect that combination in **series** with $2\,\Omega$:

$$\frac{1}{R_p} = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{3}{6} = \frac{1}{2} \implies R_p = 2\,\Omega$$

$$R_{\text{total}} = 2\,\Omega + R_p = 2\,\Omega + 2\,\Omega = \mathbf{4\,\Omega}.$$

3. **(ii) To get** $1\,\Omega$:
   - Connect all three resistors in **parallel**:

$$\frac{1}{R_p} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3+2+1}{6} = \frac{6}{6} = 1 \implies R_{\text{total}} = \mathbf{1\,\Omega}.$$

Template 3: The 4 Essential Chemical Salts (4 to 5 Marks — Q38/Q41)

Problem:

> Write the chemical formula, preparation equation, and two important uses of:
> (i) Bleaching powder
> (ii) Plaster of Paris

Complete Model Answer:

1. **Bleaching Powder**:
2. **Chemical Formula**: $\mathrm{CaOCl_2}$ (Calcium oxychloride)
3. **Preparation Equation**: Produced by the action of dry chlorine gas on dry slaked lime:

$$\mathrm{Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O}$$

4. **Two Uses**:
   1. As a disinfectant in drinking water to kill germs and make it potable.
   2. For bleaching cotton and linen in the textile industry.
5. **Plaster of Paris (POP)**:
6. **Chemical Formula**: $\mathrm{CaSO_4 \cdot \frac{1}{2}H_2O}$ (Calcium sulphate hemihydrate)

7. **Preparation Equation**: Prepared by heating gypsum ($\mathrm{CaSO_4 \cdot 2H_2O}$) at $373\ \mathrm{K}$ ($100^\circ\mathrm{C}$):

$$\mathrm{CaSO_4 \cdot 2H_2O} \xrightarrow{373\ \mathrm{K}} \mathrm{CaSO_4 \cdot \frac{1}{2}H_2O} + 1\frac{1}{2}\mathrm{H_2O}$$

8. **Two Uses**:
    1. Used by doctors as plaster for supporting fractured bones in the right position.
    2. Used for making statues, decorative casts, and smooth false ceilings.

Template 4: Structure of Nephron & Urine Formation (4 to 5 Marks — Q42)

Problem:

> Describe the structure of a nephron with a labeled diagram. Explain the three main steps involved in urine formation.

Complete Model Answer:

1. **Structure of Nephron**:
2. The nephron is the basic structural and functional filtration unit of the human kidney.

3. It consists of:

    - **Bowman's Capsule**: A cup-like sac at the beginning of a nephron.
    - **Glomerulus**: A dense network (tuft) of blood capillaries enclosed inside Bowman's capsule.
    - **Renal Tubule**: Convoluted tubule divided into proximal tubule, Loop of Henle, and distal tubule.
    - **Collecting Duct**: Collects filtered urine from several nephrons and carries it to the ureter.

4. **Three Steps of Urine Formation**:

5. **Step 1: Ultrafiltration (Glomerular Filtration)**:
    - Blood enters the glomerulus under high pressure. Water, glucose, amino acids, urea, and salts filter across the glomerulus into Bowman's capsule as the initial filtrate. Blood cells and large proteins remain in the blood.
6. **Step 2: Selective Reabsorption**:

    - As the filtrate flows along the renal tubule, essential substances like all glucose, amino acids, major amounts of water, and vital salts are selectively reabsorbed back into surrounding blood capillaries.
7. **Step 3: Tubular Secretion**:
    - Waste substances such as excess potassium ions, hydrogen ions, and drugs are secreted from the blood into the tubule fluid, forming the final urine that enters the collecting duct.

Template 5: Mendel's Monohybrid Cross & Sex Determination (4 to 5 Marks — Q40)

Problem:

> (a) Illustrate Mendel's monohybrid cross between a pure tall pea plant (TT) and a pure dwarf pea plant (tt). State the phenotypic and genotypic ratios of the F_2 generation.
> (b) Explain how the sex of a child is determined in human beings.

Complete Model Answer:

1. **Mendel's Monohybrid Cross**:
2. **Parents**: Pure Tall (TT) $\times$ Pure Dwarf (tt)
3. **Gametes**: T and t
4. F_1 **Generation**: All plants are Tt (All Tall, showing law of dominance).
5. **Selfing** $F_1 \times F_1$ **(**$Tt \times Tt$**)**:

Gametes	T	t
T	TT (Tall)	Tt (Tall)
t	Tt (Tall)	tt (Dwarf)

- F_2 **Phenotypic Ratio**: $3:1$ (3 Tall : 1 Dwarf)
- F_2 **Genotypic Ratio**: $1:2:1$ ($1\,TT:2\,Tt:1\,tt$)
- **Sex Determination in Humans**:
- Human cells contain 23 pairs of chromosomes. Pair 23 are the sex chromosomes.

- Females have two identical sex chromosomes: XX. All female eggs carry an X chromosome ($22 + X$).
- Males have two different sex chromosomes: XY. Half of the male sperms carry an X chromosome ($22 + X$), and half carry a Y chromosome ($22 + Y$).
- **Mechanism**:
  - When an X-carrying sperm fertilizes the egg $\rightarrow XX \implies$ **Female baby (Girl)**.
  - When a Y-carrying sperm fertilizes the egg $\rightarrow XY \implies$ **Male baby (Boy)**.
- **Conclusion**: There is an exact 50% probability of a male or female child, and the genetic sex of the child is determined exclusively by the father's sperm.

Template 6: Cleansing Action of Soap & Micelle Formation (3 to 4 Marks — Q37)

Problem:

> Explain the mechanism of the cleansing action of soaps. What is a micelle?

Complete Model Answer:

1. **Structure of a Soap Molecule**:
2. A soap molecule is a sodium or potassium salt of long-chain carboxylic acid (e.g., sodium stearate $\mathrm{C_{17}H_{35}COONa}$).

3. It has two chemically distinct parts:

    1. **Ionic Head ($-\mathrm{COO^-Na^+}$)**: It is polar and **hydrophilic** (water-loving). It dissolves in water.
    2. **Hydrocarbon Tail ($\mathrm{C_{17}H_{35}-}$)**: It is non-polar and **hydrophobic** (water-fearing). It dissolves in oil, grease, and dirt.

4. **Micelle Formation & Cleansing Action**:

5. Most dirt and grease on clothes are oily in nature and do not dissolve in water.
6. When soap is dissolved in water, soap molecules arrange themselves radially in spherical clusters called **Micelles**:
    - The hydrophobic tails face inwards, dissolving into the central oil/dirt droplet.
    - The hydrophilic ionic heads face outward towards the surrounding water.

7. When the cloth is scrubbed or agitated, the micelles remain suspended as an emulsion due to ion-ion repulsion between heads. The trapped dirt droplet is pulled out into the water and easily rinsed away.